PARMCO - Bluetooth Development Chat Log

October 22, 2026 | Checkpoint 2 | Raspberry Pi 4 + MIT App Inventor

Purpose

Record of the user prompts and concise summaries of the assistance and troubleshooting during Bluetooth Classic serial communication setup. User prompts are quoted; assistant responses are summarized rather than reproduced verbatim.

1. Set up Raspberry Pi Bluetooth server

User prompt: "Alright so the next thing on my end is to set up the bluetooth connection on the pi. give me the code for it"

Assistant response summary: Provided a C BlueZ RFCOMM server listening on channel 1. It receives newline-terminated START, STOP, DIR:CW, and DIR:CCW commands, prints them, and sends ACK replies. Included installation, compilation, pairing, and Serial Port Profile registration steps.

2. Initial SDP server failure

User prompt: "Failed to connect to SDP server on FF:FF:FF:00:00:00: No such file or directory"

Assistant response summary: Explained that the legacy SDP interface might be unavailable; outlined enabling bluetoothd compatibility mode, restarting Bluetooth, and registering the Serial Port service with sdptool.

3. SDP permission error

User prompt: "Failed to connect to SDP server on FF:FF:FF:00:00:00: Permission denied"

Assistant response summary: Suggested retrying SDP inspection and service registration with sudo, then checking the bluetoothd process and service status if access was still denied.

4. Confirm service registration

User prompt: "its there now"

Assistant response summary: Moved to compiling and running the RFCOMM server, pairing the Android phone, connecting through MIT App Inventor, and testing commands and reconnection.

5. Identify Pi Bluetooth name

User prompt: "What should the name of the Pi be in my bluetooth settings"

Assistant response summary: Explained how to inspect the Bluetooth Name and Alias with bluetoothctl show and optionally set an alias such as PARMCO-Pi.

6. Pi appears as an audio device

User prompt: "...so the pi seems convinced that its an audio bluetooth device and is not connecting to the server anymore"

Assistant response summary: Clarified that audio service advertisements do not preclude SPP. Recommended verifying the Serial Port service, server process, pairing, and connecting from the App Inventor app rather than Android Bluetooth settings.

7. Connected; commands rejected

User prompt: "alright we're connected. These are the outputs in the pi terminal I get when pressing buttons in the app:
PARMCO Bluetooth Server
Listening on RFCOMM channel 1
Waiting for Android connection...

Android connected: 08:8B:C8:51:4C:29
Received: Start
Sending: ERR:UNKNOWN_COMMAND
Received: Stop
Sending: ERR:UNKNOWN_COMMAND
Bluetooth receive: Connection reset by peer
Waiting for Android reconnection...

Android connected: 08:8B:C8:51:4C:29"

Assistant response summary: Diagnosed case-sensitive command matching: Start and Stop did not match START and STOP. Advised changing App Inventor SendText payloads to uppercase while retaining newline delimiters. Noted the server accepted reconnection after a reset.

8. Acknowledgments confirmed

User prompt: "it sends acknowledgements"

Assistant response summary: Confirmed successful bidirectional app-to-Pi communication, command acknowledgment, and reconnection behavior; recommended saving the C source, .aia project, and evidence screenshots.

9. Request documentation

User prompt: "alright so give me a pdf for my prompts for today for me to upload to github for documentation"

Assistant response summary: Prepared this prompt-and-response summary for GitHub documentation.

Outcome

Android paired and connected to the Raspberry Pi RFCOMM server. The Pi received commands and sent acknowledgments that the Android app displayed. The command capitalization mismatch was identified and corrected. Reconnection was observed after the client connection reset.

Scope: This document records Bluetooth communication work, not stepper motor actuation.